

what the words leave out

the channel is lossy in both directions — measured in RepQA

So: connect brain signals without requiring verbal instructions

PART 1 The Foundation
Models for Brain Signals

HyBRiD

Neuro-attn

FA-EEG

ML → prior

prior → ML

neural priors — how the brain is organised

the prior travels back out

SWA

signals → semantics

PART 2 The Bridge
From Signals to Representations

PRISM I
Choose alignment space

MindLLM
Align fMRI to language

Language representation
Read out multiple brain functions

PRISM II
Language-to-vision
Object-centric generation

D-Edit
Item prompts + attention
Stronger disentanglement

semantics → action

PART 3 The Embodiment
From Representations to Interactions

MWM

Brain signals
to action meaning

EEG latent actions
predict visual consequences

Act toward specified goals

PHIL
Learned heuristic
for graph search

DySta
Visual policy
Memory + fast inference

Separately evaluated capabilities

brain signals, semantics, and behavior — complementary research capabilities